

Power-Limited Winch Design

Design a winch that lifts the greatest useful load without exceeding a motor or power limit.

Advanced	2-3 class periods	Engineering Challenge
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Engineering Context

A drone recovery station has a small motor but must lift equipment safely. Your team must balance drum radius, speed, torque, and structural support.

What You Will Practice

- Relate torque, radius, force, speed, and power
- Use calculations to guide design
- Test a mechanism under load
- Evaluate design tradeoffs

What You Need

- Low-voltage motor or hand-crank simulation
- Spools of different radii
- String
- Masses
- Timer
- Structural materials
- Engineering notebook

Student Instructions

- 1 Record the available motor speed or hand-input condition.
- 2 Create at least three drum-radius concepts.
- 3 Predict which radius will lift fastest and which will lift the greatest load.
- 4 Build a supported winch with one selected drum.
- 5 Test a standard load and record lift time.
- 6 Increase load in approved increments until the design reaches the limit.
- 7 Calculate output work and average power for one successful lift.
- 8 Revise the drum or support and repeat the test.

Engineering Requirements

- Lift through a fixed height
- No uncontrolled dropping or slipping
- Include a stable axle support and secure string attachment
- Use at least one calculation before selecting the final drum
- Final design must complete three repeatable lifts

Constraints & Safety

- Use instructor-approved voltage and masses
- Keep hands clear of winding string
- Do not stall a motor for extended time
- Use a catch zone below the load

What You Submit

- Concept comparison
- Torque/radius calculations
- Test data
- Final winch photo or CAD
- Design tradeoff explanation

Reflection Questions

- 1 How did drum radius affect speed and lifting force?
- 2 What limited the design first: motor, structure, traction, or string?
- 3 Where was power lost?
- 4 What safety factor would be appropriate for real cargo?

Engineering Tip A smaller drum increases available line force but requires more rotations to lift the same distance.	Common Mistake Calculating torque without using consistent units is a common source of error.
Stretch Goal Add a ratchet or brake that prevents rollback during a power loss.	Career Connection Actuation engineers design winches, reels, and deployment systems for aircraft recovery equipment and spacecraft mechanisms.